

Active Inference Institute ~ 2023 Quarterly Roundtable #4

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Duration: 44:28 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello and welcome everyone it's December 8th 2023 and we're here in 2023 quarterly Round Table number four at the Active inference Institute let us do it Welcome to The otin Institute we're a participatory online Institute that is communicating learning and practicing applied active inference you can find us at links on this slide this is a recorded and an archived live stream so please provide us with feedback so we can improve our work all backgrounds and perspectives are welcome and we'll follow video etiquette for live stream today in the quarterly round table that closes 23 we'll go over first The Institute scale then the organizational unit scale with edu active and Reiner and finally close with some thoughts on 24 so before we go any further Alex and blue do either of you want to say hi or share anything or just give any overall thoughts that you don't see coming up later on yeah I mean just that the year was just so epic and um really like we're still climbing we're still in the exponential growth phase of our uh Colony so it's um been amazing and awesome to work with you guys and The Institute and just watch it grow and develop and all the new things that have bubbled up it's been awesome thank you blue Alex yeah maybe some idea but from current position we are seeing and considering like now a scale of ecosystem scale and uh like more broad field more more and more people uh educational institutions and uh startups actually last time everything is developing involving and we are in a good place thank you all right so either of you please just raise your hand if you want to add anything on a slide or we'll move on so first to some updates at the organizational scale well it was just in August of this year when with many co-authors we brought together this document on the active inference Institute and the ecosystem and figure one is still operative and totally in effect we'll be talking more about that as we go but during this year we did Reach This Milestone or this checkpoint of being able to even assemble the team and the work this way so this was a big milestone for us and also gives us something like a map for the next phase I'd like to thank our first cohort of the board of directors Raph calman John clippinger myself Dean tickles blue Knight and Mike Smith

great work as well as to the first cohort of officers us here and the second cohort of the scientific Advisory Board listed here also just a broader and grander and more diffusive thanks to everybody who's engaging if you're listening to it you are so it wouldn't be the way it is without everything being the way it is so it's been a great year and we're really excited for what's to come if you're interested up until December 15th we will be accepting applications for the scientific Advisory Board it's not an active inference test it's about people who are beginner to expert in active inference who are looking to make a contribution to the Institute in this scientific and advisory capacity it would be awesome to see an Engaged diverse and Global Group as during 2024 we expect to bring the Sab through on from where it was to how it will be next and that's going to be very exciting differentiating different channels of participating an Sab memb is not just somebody who is joining a textbook group or not just somebody who's facilitating project so what is it that differentiates that that's what we want to explore if you want to be part of that work in this phase then go to the website here and check for more information about our nonprofit status well last round table we updated that we had submitted our form 1023 to the USA IRS so our tax exempt status as a 501c3 is pending in the last few weeks we've we set up a bank account so we now can actually receive Fiat Financial donations if you're interested in just spontaneously feeling like you'd like to donate right now reach out otherwise we'll be developing and releasing more information on our philanthropic development during the year and this is a special call for anybody who has experience or networks or desire or feels like they have capacity to contribute to a kind of fundraising program in some way get in touch this is the right time to do it about Institute scale affordances we continued with our volunteering and internship programs volunteering and engaging with projects is going to be welldeveloped in 2024 we'll come back to that later about the internship we had overall during 2023 about 30 interns who were active to different stents and six that have successfully completed and a handful that are continuing on the internship has been truly amazing I could say many many things about the wonderful and unique work that people did it's been some of the finest and most fun work about the broader information environments and our footprint and phone distribution there here's just some numbers from different social media platforms and different channels any more Institute scale updates from either of you or we'll go on to the organizational units okay on edu active the organizational unit with a focus on education we continued activity in the ACM textbook group group we completed cohort 4 so great work to everybody who stuck on through and joined in throughout all the videos are available for all the completed cohorts cohort 5 again a

very creative and inspiring and ambitious group of participants who got through chapters 1 through five over the previous months they'll be heading into part two in 2024 and in 2024 we'll release more information on this soon it'll be really exciting to continue the types of value adding epistemic services that we've templated over the previous one and a half years in the textbook group related to the integration with ontology and math learning generative AI and so on and also brooch out into more groups different artifacts and also non-english languages blue yeah I just wanted to add um that this is also an open invitation for people who have experience would like to learn more would like to scaffold various educational components related to active inference or any of the foundational skills like please get in touch with us or if you have ideas of a course you would like to see how can we make it happen together um I I really would just like to hear from the broader Community with what what they want and need so thank you blue as with active inference two ways to resolve the differential change your mind or change the world if you want a translation of this into that or a more accessible introduction to this or that changing the mind is just coming to terms with a fact it doesn't exist and resolving the differential that way or taking the action to do it if it's within your capacities and so hopefully we have a setting where both those can happen blue oh still raised on my side but great um this year we completed two courses we had zero completed at the end of last year and so that was a major component of our year and works and just in the previous months there was the completion of the course led by Chris Fields physics as information processing it was a truly unique and very groundbreaking course and body of work presented some of the next steps here are to curate and publish the 2023 proceedings as an open source artifact and then consider where we're at with future learning cohorts and project-based work in Quantum active inference free energy principle and a lot of other areas anyone want to add anything more on the Physics course and the second course that was complete completed over the last few months was constructing cultural landscapes active inference for the social sciences with AEL Ben ma Lena and myself this was another fun Journey um similar next steps in terms of publishing the transcripts and the proceedings and similarly considering what people are interested in doing and engaging with about active inference in the social sciences setting a topic that many people would I think love to continue working in so don't be shy if you want to continue that work outside of the courses here's the live streams that we had since Round Table Three on September 29th 2023 we completed the physics and social sciences courses as mentioned we had an or stream with Anam marce one we had two paper base live stream series with a0o background and context and a one and A2

with authors and that was live stream 55 with Magnus kudal on the realizing synthetic active inference agents message passing work and live stream 56 with blue and with Alexander Aria on the cognitive modeling and Mortal computation both were just incredible Rich epistemic resources and we had 10 guest streams over that time covering topics ranging from AI Safety and Security thermodynamic Ai and World models neural Linguistics philosophy symbology architecture really fun panel of guest streams and we plan to do an end of December review stream possibly formatted as a comedy so whether you think that you'd like to joke or just laugh let us know if you're interested in that also we introduced a new series which was the active inference insights podcast series that was led by Darius parvy Wayne and with blue facilitating this was a informal insightful and inspirational conversational setting at least as I saw it um the four that have already been lined up or delivered were number one with Carl friston two with Mal abason insights 3 with John Veri just premiered yesterday and next week iness hippoo in insights 4 and many many more to come so not not to Ros Darius here but I think the insights are loading as Carl is speaking blue uh yeah this has been awesome and um just a big shout out to to Darius who's super motivated and I'm really pushing hard on this uh right now this is up on YouTube and episodes are coming out bi-weekly as just an audio podcast right now on Spotify Amazon music we're hoping to load up on um Google and apple podcasts soon so it just takes a little time for the stamp of approval to to come through so great and following all audiovisual production we pass off the work to the journal where we are with the work of Alex Holly Dave and others are working on the continuous deployment and development of live stream and AV transcription curation and publication this is a great infrastructural project and so it continues to release curated transcripts which promote the accessibility of these streams otherwise it can feel very ephemeral with the audio content or hard to search through or hard to parse but by getting things into a plain text format we're going to be able to do incredible things with translations captions accessibility language models and more those are the main pieces from edu active in Reiner we have a little bit less to say we have the continuation of the works that have been happening and there was a blizzard of research papers coming out all over the place including with many Institute participants and one of our projects in 24 on knowledge engineering will be how to effectively curate the research that is coming out as well as hosting research at The Institute so that we can have a summary and be able to come back to you for a future round table and just have a nice Grid or list of what works came out across different fields Fields across different topics different languages and so on okay on to the fun of where we go in 24 so Alex and blue all all

look forward to your thoughts as we continue here so first just speaking to the Institute scale of where we're going in 2024 here we have the figure one from our ecosystem paper and some version and continuations are shown here in 2024 we are going to have the second cohorts of our board of directors second cohorts of our officers these will be all the same people continuing on we will have the third cohort of our scientific Advisory Board as discussed a little bit earlier and we'll have two 1.0 that's going to be the organizational unit lead who's the blanket or the interface between the Institute scale and the project scale in the edac education unit and in the re inference Research Unit and we're also due to the recently mentioned capacities be engaging in a philanthropic and Grant application and hopefully receipt 1.0 so these are some of the upstairs changes at the Institute anything blue or Alex you want to add or just like that's exciting or anything here cool carrying on then to the project scale projects are where the work is these are where the rubbers are hitting the roads on the left is an image from Chris fields's Quantum active inference work and briefly we have a classical screen with two systems that are communicating across the holographic screen with a class classically inscribed information so we have Quantum Alice and Quantum Bob communicating through this interface so aligned with our roots in active inference system engineering open science and cognitive security we will be implementing a multiscale temporal architecture for Project Life Cycle management bringing out pieces that we didn't have in 23 and 24 as we will hopefully for years and years to come we'll be considering projects as active or inactive at the quarterly scale not unlike a class but also not like a class but also like a class projects will be proposed for activity status by a facilitator who takes that responsibility and accountability before the quarter this corresponds to preparation so Quantum measurement preparation we'll provide a form that will ask about the structure function and impacts of the project how does this project contribute to the accessibility of active inference how does this project contribute to the rigor of active inference how does this project contribute to the applicability of active inference and so on just some questions for people to start thinking about we will review those proposals to determine if they're prepar to make that measurements at The Institute projects will convene and be carried out through the quarter based upon what resources they do or don't need from The Institute maybe a synchronous meeting link maybe a Discord Channel maybe something else and at the end of the quarter there will be the measurement phase prepare measure prepare measure and in that measurement we'll be looking to hear back from the projects ask in something like the shadow of what we asked them to prepare for how did your work contribute to accessibility and so on that will constitute the measurement then we

will curate those measurements for you all and present them at the round table and discuss them with the scientific Advisory Board where we'll then make assessments and reallocations we hope to continue to scaffold the efficacy and efficiency of faster time scales of inference attention and learning on through slower lifelong and intergenerational scales this is just about the timing of the projects so let's go into what those project types are or will be in 24 and then BL Alex please feel free to make some comments so we'll be separating things out into broadly three categories of projects first category of project are going to be in Institute core projects these are infrastructural and open science ecosystem type services such as the active inference ontology translations and applications production audiovisual production with a live streams and podcasts active inference Journal reproduction transcriptions and translations digests this will be grappling with a volume of Publications and different form and active inference and also developing some Communications channels that people can opt into at for example different levels of technical depth and different languages active blocker active inference generative modeling using cat cat knowledge engineering supporting the digest work educational activities qualifications course development groups and so on service and governance activities and communication activities so these are CORE Institute projects and by separating out those projects we'll be able to have a little bit more clarity about this middle category which just tentatively are considered as Institute hosted projects so it's a project hosted at The Institute these projects are nested under either edu active or Reiner whether they are more of an educational project or research project and there's going to be a few different categories or types Within here there will be some topical and working group projects so these are really the system of Interest or disciplinary or subject matter type groups such as health robotics Linguistics spatial web meta governance philosophy entomology and so on so if you're super excited about active inference and X this is that group join it if it's there and have fun and if it's not there again come to terms with the fact that it's not there or be the one who makes that preparation here also we'll have projects like sanjie num J's textbook in progress and reviewing as an educational hosted project as well as research project so if somebody wants to have their research project hosted at The Institute then that's the space for that and apart from the subject matter and system of type projects we will open up an orthogonal dimension of projects for skills and practices so this might include again depending on who prepares for what this could include computational modeling and programming beginners experts mixed levels All the Above reproducibility This was a really fun conversation with Andrew P where we talked about a reproducibility working group where a paper comes out we

investigate that paper we really analyze it understand it run the code replicate the figures that would help us understand how is active inference being applied what does it really look like to get to that paper being written and then we've basically de facto done a zero so at that point we could then contact the authors if we hadn't already have them for conversations on live streams or podcasts also skill and practice groups like math beginners and math advanced math art and creativity and attention management various other skills we will be sharing information through our newsletter channels and maybe with a short video and so on about how you can propose to lead a hosted project following that we will be providing information on how you could join those projects that will be either active in the core of the Institute or hosted in The Institute in 24 and again it's quarter by quarter so if it's not the right quarter for you to begin in first quarter of 2024 prepare for the second quarter if that's not right prepare for the third quarter but everyone can be involved in these projects whether or not they've proposed to lead them and hopefully this structure with a quarterly time frame will help make it clear who's leading a project what's the project about what is in or out and then there are the ecosystem projects everything happening in the active inference ecosystem which goes way way way Beyond The Institute here are all of the things that people are doing that aren't included in the core projects or hosted projects and for projects in this space whether it's your startup or your open science initiative we can support through hosting your participation and engagement in learning groups or with other resources that can be provided and exploring Partnerships and collaborations and so on so this structure for the projects in 2024 has been arising over the last several days it's just presented as a sketch and a direction if you're watching this video in realish time semi- sync time then consider it and maybe you're interested to lead a project or maybe just see where this all goes blue or Alex want to add anything on the projects um possibly yes want to add about but for now we have that uh basic nested structure and it's aligned with inference approach systems approach and uh using it we are ready for scaling activities and support them uh as organization and this kind of uh hosting projects for now all projects was uh in initiated from Institute possibly next year we'll see more projects uh which are born in some our communities or somewhere but finally supported and hosted in Institute as some kind of platform and service for ecosystem thank you I think that those life cycles and those life histories will be really exciting like started in a conversation hosted and incubated in The Institute for these two quarters and then out and Beyond or part of it moved into a core infrastructural service functionality or and any and all blue any other thoughts on the projects it's all so exciting like when when can we start where do I sign where we start I know how do I

sign up how can I get involved um um uh just good stuff good stuff thank you blue yeah um we're just trying to summarize and have some concise fun here and recap and wish everyone a very restful end to this year and beginning to the next if anyone's watching live please feel free to write any questions into the chat the rest of this stream I will give a brief closing address then we'll look at any comments in the live chat then Alex and blue any other comments that you want to have so to make a brief closing presidential address to give the takeaway first thank you for your attention and upcoming action let's be who and how we are can't do it any other way wouldn't want it any other way this is just to give a little bit of some timeline as now we are completing our third year of activities in The Institute and I just wanted to share a little bit about that timeline not in an overly personal fashion but in a way that hopefully provides some information to the kind of person who's 28 minutes and 35 seconds into this video 2020 was like the do zero for the work that we're exploring here the background and context this is when Alex and I met I had already known blue and in work with RJ and cogc we explored and researched the preset what does it mean to preet especially in the context of remote teams and operational art we started act in lab as it was then known 2021 we formed stormed normed and performed a lab and we saw the incipient structuring like an imaginal dis of edio active and re inference in their manifestations as doedu andtools more like subdomains of actim and in CAC we worked on narrative information ecosystems in 2022 22222 222 we had the lab in a nutshell in the sense of Hamlet like I could be bound in a lab and count myself a king of Infinite Space and we left that nutshell that Acorn and the lab transformed into an Institute we also had some structural morph logical organizational development as our first scientific Advisory Board came into existence as an informal informal unconventional Advisory Board and in cognitive security we worked on structuring the information comments 2023 our year is now coming towards a conclusion having become an Institute during 2023 was something like addressing the question now that we're in Institute what next well we had our first board of directors and officer cohorts that dove into bolts business operating legal Technical and social and we developed out the edu active and the Reiner aspects of our work with a special focus on the edac active as this earlier presentation speaks to as well and on a more technological note we left behind active inference gmail.com we still have the logins don't worry about it but we've moved things to the active inference do Institute workplace so that's where we now have personal emails and so on so that's really a nice niche and information environment to build in there were many unities as plural and minimum to as well as unities as plural and minimum four this year in the minimum 2 domain I would Place one of

our key artifacts which was The Institute and ecosystem as this agent Niche enclosure and we explored that in the third applied active inference Symposium on enacting ecosystems of shared intelligence that was a really fun participatory organizing experience and the panel and the conversations and the presentations were just incredible too there were other Min twos and Min fours ranging from different topics and how they were brought together for example Deans classic tetrad active inference abduction category Theory and dissemination as identity and there were some other mints and Min fors that anyone who wants to speak more about them can talk another day but I'll just leave them posted here and in our cogc work we pursued and yesterday published The Atlas specification for utilizing Dynamic knowledge bases of pattern languages that are navigated through questions which brings us to 2024 who what why where when how from here at the end of 23 and into through Beyond 24 where will you and we go from here so any general comments or any other thoughts from Alex or blue and then I'll look at any questions in the live chat you know as we learned somewhere in some stream or some course that the future is just an of a memory so where will we go from here we really already went do you remember where you went or where you will go from here I can't say I do but I know what you're saying okay I'll ask Alex please first and then I'll look to the live chat yeah maybe it's what what wasn't so highlighted uh during this uh this round table but um very important achievements on educational side and actually possibly the main change from the lab into Institute that now we have uh uh hundreds of people which are somehow learned uh something about active inference in different formats and now physics that's actually great and uh that people definitely will share share uh like epistemic status for active inference for ours and that's really great thank you thank you Alex thank you I agree with this okay we'll have time for as many questions are in the live chat as people ask so first Andrew is the act inference Institute pursuing grants or funding if so how is it prioritizing where it wants to apply monetary resources to the first question are we pursuing grants or funding yes for both our first major grant that we applied for was submitted in September of this year for the NSF pose program which was about Pathways to sustainability for open source ecosystems we don't know how that plays out we close the year in a Quantum uncertainty there but yes we do plan to continue pursuing grants of various kinds we've also already been a collaborator and providing letters of support to other people in their grants and so we can have conversations with people who are wanting in their own country or in their own location to apply for a grant to them and for their broader impacts maybe they want to commit to organizing a project or to hosting some other kind of activity at The Institute to make a win-win proposition for us all so

yes about the funding and the private philanthropy angle we were not able to do that in 2023 we will be doing that in 2024 now that we are set up with a bank account and submitted our form 1023 and so on so yes we hope to have any and all channels for um monetary inflows grants and private philanthropy and then to the second question how are we prioritizing where we want to apply monetary resources that's a bigger question of course I can't give a complete answer but all of us are super active in diligent on the financial and all bolts processes so I hope we'll have an effective and efficient way to allocate our attention and other kinds of resources and we're gonna do it right blue um so just uh one of the major scaffolding infrastructure things that we're building in 24 flushing out in 24 is like our financial policies procedures investment procurement all of these things um and so that will be made clear as we go forward through this year um and then also prioritizing like where do you spend the money frequently when you're applying for Grants like you're apping to spend the money in a certain way um and so like there are lots of projects that are as we just saw on the list that are scaffolded partly through or hosted by The Institute and like we are happy to um also facilitate and scaffold the seeking of funding for these projects um so as we go through like we will prioritize the projects that have funding because they have funding and we will help people get the funding if they want to um do that or we will we will scaffold that that endeavor to be funded to the best of our ability thank you blue yes indeed if after some successful quarters of operation a project wants to lead a Grant application and it goes to them and they just need the letter of support that's one kind of archetype another method might be we identify a grant opportunity at The Institute scale and then pass that down and say which of these projects can we include at x amount and there will be step ways into active inference professionalism with and through the Institute and with and through the ecosystem so many paths just be on the journey okay Susan asks can you say more about developing the pattern language Library which is atlas sure I'll just speak briefly to how it connects to active inference Atlas is a flexible pattern language knowledge database structure that humans and other synthetic intelligences can navigate with questions it allows some really interesting ways to work with different kinds of information patterns potentially all but at the very least some for example cognitive patterns will have active inference implications and active inference specifications or representations in that way the composability of active inference models and the composability of patterns will be finding a new Integrity between each other we'll have patterns about generative models oh this is a nested model this is a deep temporal model this is sophisticated active inference this is and so on and then we'll use active inference for patterns

that aren't necessarily active inference patterns overconfidence bias and so on and be able to develop motifs that will lead to some pretty new and exciting ways to work with information upcycle Club rights is it possible to collaborate with other authors on joint research papers that arise and are related to the aii course taught by Chris Fields yeah great question so first just to address it generally is it possible to collaborate with other authors on joint research papers yes within the limits of plausibility and consent and all of that absolutely about the physics as information course by Chris Fields yeah that is like absolutely what we would love to see happen and just to give an example of where that could go first off anyone who wants to continue their work in the active inference ecosystem leveraging what they learned in the physics information processing course please continue now if you'd like to potentially host that at the Institute then when we have the form for proposing a project then it can be filled out I'd like to lead an exploratory research working group on this and this here's how I see some possibilities of how it might apply and how it might help active inference ecosystem in this way it's going to be open science along these ways of thinking about it and here's what I'll need a weekly video link a Discord Channel or whatever it happens to be and then yes we would love to have somebody lead the charge on that submit that form and then create that attractor so that people who want to participate in that research can then go there that group could then decide yeah we're gonna um summarize our notes and publish a pre-print at the end of every quarter so if you join along for 10 weeks of activity then you're a co-author at the end of the quarter or the groups can do whatever they want they can choose to structure things how they want and then just report on it in the measurements we'll talk about it with future use and other use on the round table return to it with a more comedic lens at the end of the year and hopefully this will be like a really productive and inclusive way to do science and have fun in our ecosystem if anyone else has a comment in live chat just a few more seconds otherwise Alex or blue what are your closing comments just thanks everyone for your attention and work and continued persistence and study of active inference it's good it's a good field to operate within and Beyond and the nice scaffold to have Alex anything in closing as usual we're in the beginning of infinity so let's keep walking thank you great times thank you also everyone for the comments in the live chat hope to see everybody in the Discord just join the voice chat be the first one and scaffold it for somebody else to join in reach out on the edges stay tuned for where we go in 24 acting first surf